Shrikaran P

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Professional Summary

Computer Science Engineering student with a strong foundation in Java, Python, Data Structures & Algorithms, Object-Oriented Programming, and SQL. Hands-on experience in software development, backend application development, REST APIs, databases, and cloud fundamentals through academic, research, and real-world projects. Strong analytical and problem-solving skills with a passion for building scalable, reliable, and high-quality software solutions.

Technical Skills

- **Programming Languages:** Python, Java, C
- **Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming (OOP)
- **Backend & APIs:** FastAPI, REST APIs
- **Data Engineering:** ETL Pipelines, Apache Spark (PySpark), Data Warehousing, Apache Airflow, Apache Kafka, Data Ingestion, Data Validation
- **Databases:** SQL, MySQL, PostgreSQL
- **Web Technologies:** HTML, CSS, Javascript
- **Software Engineering:** Software Testing, Debugging, Agile/Scrum
- **Version Control:** Git, GitHub
- **Cloud & Tools:** AWS, Firebase, Google Colab
- **AI/ML:** Machine Learning, Deep Learning, PyTorch, Scikit-learn, CNN-LSTM
- **Soft Skills:** Problem Solving, Team Collaboration, Communication, Leadership, Quick Learner, Time Management

Educational Qualifications

B.Tech in Computer Science Engineering(6th semester) May 2023 - May 2027 SRM Institute of Science and Technology - Tiruchchirappalli
CGPA - 9.3/10

Achievements

- **AWS Academy Cloud Foundations – AWS –** Learned fundamentals of cloud computing and data storage for scalable data processing
- **Oracle Cloud Infrastructure (OCI) AI Foundations Associate** – Earned Oracle certification demonstrating foundational knowledge of AI, Machine Learning, Generative AI, and Oracle Cloud services.
- **3rd Prize Winner (Cash Award)** – Ideathon, SRM Institute of Science and Technology, for developing a Centralized Dam Management System.
- **Participant** – 24-Hour Hackathon, Dhanalakshmi Srinivasan University, where I developed a Farmer Market Price Prediction System as part of a team.
- **Foundations of AI/ML – GeeksforGeeks -** Gained hands-on experience in data preprocessing, model building, and evaluation techniques
- **Java Full Stack Development – EduSkill** - Built web-based applications with structured data handling and backend integration

Research Internship(NIT Tiruchirappalli)

- Motor Imagery EEG Classification (BCI) — Worked with BCI Competition IV Dataset 2a and BCI Competition IV Dataset 2b EEG .gdf recordings for motor imagery analysis.
- Preprocessed raw EEG signals, selected motor cortex channels, and generated STFT-based time–frequency spectrogram representations.
- Developed CNN-based models for left/right hand motor imagery classification and subject-wise performance evaluation.

Research & Publications

- A Secure On-Premises ETL Pipeline Using OCR and Local LLMs - ICCMC 2025, Tiruppur
- Contrastive Knowledge Graph-RAG for Concept Misconceptions-RECCAP 2026, IIT Palakkad

Projects

Smart Academic Timetable Generator

- Developed an automated timetable scheduling system that generates conflict-free academic timetables by allocating subjects, faculty, classrooms, and time slots while satisfying institutional constraints.
- Implemented scheduling logic and optimized resource allocation to reduce manual effort, improve timetable accuracy, and enhance administrative efficiency.

Weather Data Pipeline API

- Built a FastAPI-based REST API to ingest real-time weather data from the OpenWeather API, process it, and store it in PostgreSQL for efficient data management.
- Implemented secure REST API endpoints with request validation, filtering, and historical weather retrieval, enabling efficient data access and analytics.

A Secure On-Premises ETL Pipeline Using OCR and Local LLMs

- Designed a secure ETL pipeline to extract text from scanned documents using OCR and transform unstructured documents into structured datasets for downstream analytics.
- Automated document processing, data validation, and information extraction while ensuring data privacy and improving processing efficiency.

Farmer Market Price Prediction System

- Developed a machine learning model using Python and Scikit-learn to predict crop market prices from historical mandi data, seasonal trends, weather conditions, and demand–supply patterns.
- Provided predictive analytics and market insights to support informed selling decisions and improve farmers' profitability.

Intrusion Detection System using CICIDS2017

- Designed a deep learning–based intrusion detection model to classify malicious and benign network traffic using the CICIDS2017 dataset.
- Compared model performance against traditional rule-based methods using accuracy, precision, recall, and F1-score metrics.

Contrastive Knowledge Graph-RAG for Concept Misconceptions

- Engineered a Knowledge Graph-based Retrieval-Augmented Generation (KG-RAG) system using contrastive learning to detect and explain concept misconceptions.
- Integrated embeddings, vector retrieval, and Large Language Models to improve retrieval accuracy, contextual understanding, and explainable AI responses.

Hostel Outing Permission & Approval System

- Developed a role-based web application to digitize hostel outing requests, approvals, parent verification, and gate verification for students, HODs, wardens, and watchmen.
- Implemented FastAPI, React, SQLAlchemy, SQLite, JWT authentication, and role-based authorization, with outing history and late-return detection.